[536709] Stochastic Processes

Yuki Chino

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5 Brownian motion

5.1 Review: Gaussian

Gaussian random variable For $a, z \in \mathbb{R}^d$, we denote by $\langle a, z\rangle$ the scalar product of a and z, i.e.

$$\langle a, z\rangle = \sum_{i=1}^{d} z_i\, a_i \tag{5.1}$$

Definition 5.1 (Gaussian random variable)**.** If the characteristic function $\varphi_X(z)$ of $X = (X_1, X_2, \cdots, X_d)$ can be expressed by

$$\varphi_X(z) = \exp\Big(i\langle z, a\rangle - \frac{1}{2} z\, V\, z^{\mathsf{T}}\Big) \qquad z = (z_1, z_2, \cdots, z_d) \in \mathbb{R}^d \tag{5.2}$$

where $a \in \mathbb{R}^d$ and $V = (v_{ij})_{i,j=1,2,\cdots,d}$, then X is called d-dimensional **Gaussian random variable**. When $d = 1$, it is simply called **Gaussian**.

If V is positive definite, the distribution can be written by density function g:

$$g(x) = \big((2\pi)^d \det V\big)^{-1/2} \exp\Big(-\frac{1}{2}\big\langle (x-a)\, V^{-1}, \tag{5.3}$$

By differentiating the characteristic function $\varphi_X(z)$,

$$a_k = \mathbb{E}[X_k], \qquad v_{ij} = \mathrm{Cov}(X_i, X_j) \tag{5.4}$$

a is called **mean vector**, V is a **covariance matrix** and we denote by $N(a, V)$ the Gaussian.

Proposition 5.2. *For $X = (X_1, X_2, \cdots, X_d)$, the necessary and sufficient condition for the independence of $X_1, X_2, \cdots, X_d$ is*

$$\mathrm{Cov}(X_i, X_j) = 0 \qquad i \neq j, \tag{5.5}$$

that is, V is a diagonal matrix.

When the covariance of X and Y is zero, we say that X and Y have **no correlation**. If X and Y are independent, then they have no correlation.

Gaussian process For $X = (X_1, X_2, \cdots, X_d)$, $a \in \mathbb{R}^d$ and d-dimensional symmetric positive semi-definite matrix $V = (v_{ij})_{i,j=1,2,\cdots,d}$, X is called d-dimensional **Gaussian** random variable when the characteristic function $\varphi_X(z)$ is expressed by

$$\varphi_X(z) = \exp\left(i\langle z, a\rangle - \frac{1}{2} z V z^{\mathsf{T}}\right), z = (z_1, z_2, \cdots, z_d) \in \mathbb{R}^d, \tag{5.6}$$

where $\langle z, a\rangle = \sum_{i=1}^d z_i a_i$ and z^{T} is a transpose of z. By differentiating the characteristic function $\varphi_X(z)$,

$$a_k = \mathbb{E}[X_k], \qquad v_{ij} = \mathrm{Cov}(X_i, X_j) \tag{5.7}$$

a is called **mean vector**, V is a **covariance matrix** and we denote by $N(a, V)$ the Gaussian.

Definition 5.3 (Gaussian process)**.** For any n and $t_1, t_2, \cdots, t_n$, if the distribution of $X = (X_{t_1}, X_{t_2}, \cdots, X_{t_n})$ is dn-dimensional Gaussian distribution, then d-dimensional stochastic process $\{X_t\}_{t\in I}$ is called (d-dimensional) **Gaussian process**.

Proposition 5.4. *Let $Y = (Y_1, Y_2, \cdots, Y_d)$ be a d-dimensional Gaussian random variables and $\mathbf{X} = \{X_t\}_{t\in I}$ be a Gaussian process. Then the necessary and sufficient condition for independence of Y and $\mathbf{X}$ is for any $t \in I$,*

$$\mathrm{Cov}(Y_k, X_t) = 0 \qquad k = 1, 2, \cdots d \tag{5.8}$$

Let $\mathbf{X} = (X_t)_{t\geq 0}$ be a Gaussian process. Then the distribution of $\mathbf{X}$ is determined by the mean $a(t) := \mathbb{E}[X_t]$ and $\Gamma(t, s) := \mathrm{Cov}(X_t, X_s)$. If $a(t)$ is a constant and $\Gamma(t, s)$ is a function of $t - s$, then the Gaussian process becomes a stationary process.

5.2 Construction of Brownian Motion

Let ξ_n $(n = 0, 1, \cdots)$ be an independent random variables identically distributed as $\mathcal{N}(0, 1)$. We set

$$X_t := \frac{t}{\sqrt{\pi}}\xi_0 + \sqrt{\frac{2}{\pi}}\sum_{n=1}^{\infty}\frac{\sin(nt)}{n}\xi_n \qquad t \in [0, \pi] \tag{5.9}$$

Then since $(X_{t_1}, X_{t_2}, \cdots, X_{t_\ell})$ is a ℓ-dimensional Gaussian random variable, $\mathbf{X}$ is a Gaussian process with 0 mean and covariance is given by

$$\Gamma(s, t) := \mathrm{Cov}(X_s, X_t) = \frac{st}{\pi} + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(ns)\ \sin(nt)}{n^2} = \min(s, t) \tag{5.10}$$

Indeed, fix s and set $f(t) = \min(s, t)$ $(t \in [0, \pi])$. Since $f(0) = 0$, we can extend as

$$f(t) = \sum_{n=1}^{\infty} a_n\ \sin(nt) \tag{5.11}$$

Computing a_n, we can obtain

$$f(t) = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(ns)\ \sin(nt)}{n^2} - \frac{2s}{\pi}\sum_{n=1}^{\infty}\frac{\cos(n\pi)\ \sin(nt)}{n} \tag{5.12}$$

On the other hand, we consider the Fourier expansion of $h(t) = t$ and then obtain

$$t = -2\sum_{n=1}^{\infty}\frac{\cos(n\pi)\ \sin(nt)}{n} \tag{5.13}$$

which shows (5.10). For $\theta < s < t$,

$$\mathrm{Cov}(X_t - X_s, X_\theta) = \Gamma(t,\theta) - \Gamma(s,\theta) = 0 \tag{5.14}$$

Therefore, by Proposition 5.4, the increment $X_t - X_s$ is independent of $\{X_\theta\}_{\theta\le s}$, and since

$$\mathbb{E}[|X_t - X_s|^4] = 3|t-s|^2 \tag{5.15}$$

$\mathbf{X}$ is a continuous stochastic process. The process $\mathbf{X}$ is called **Brownian motion** on $[0,\pi]$.

Next we construct the Brownian motion on $[0,\infty)$. Let $\mathbf{X}^{(n)}$ $(n = 0,1,2,\cdots)$ be independent Brownian motions on $[0,\pi]$. We connect those Brownian motion and make the Brownian motion on $[0,\infty)$:

$$B_t := \begin{cases} X_t^{(0)} & 0 \le t \le \pi, \\ X_\pi^{(0)} + X_{t-\pi}^{(1)} & \pi < t \le 2\pi, \\ \cdots & \\ X_\pi^{(0)} + \cdots + X_\pi^{(k-1)} + X_{t-k\pi}^{(k)} & k\pi < t \le (k+1)\pi. \end{cases} \tag{5.16}$$

Then B_t is a Gaussian process satisfying

$$\mathbb{E}[B_t] = 0, \qquad \mathrm{Cov}(B_t, B_s) = \min(t,s) \tag{5.17}$$

and $B_t - B_s$ is independent of $\{B_\theta\}_{\theta\le s}$.

Question 5.5. Prove that

$$\mathbb{E}[B_t] = 0, \qquad \mathrm{Cov}(B_t, B_s) = \min(t,s) \tag{5.18}$$

Brownian bridge Let $\mathbf{X}$ be a Brownian motion. Set

$$\xi(t) = X_t - tX_1 \qquad t \in [0,1] \tag{5.19}$$

Then $\xi(0) = \xi(1) = 0$ and $\xi = (\xi(t))_{t\in[0,1]}$ is a continuous Gaussian process with $\mathbb{E}[\xi(t)] = 0$. For $0 < s < t < 1$,

$$\mathbb{E}[\xi(s)\,\xi(t)] = s(1-t) \tag{5.20}$$

On (t,x)-plane, $(t,\xi(t))$ is called **Brownian bridge**, which is an interpolation between $(0,0)$ and $(1,0)$ by Brownian motion. If we set

$$\eta(t) = (1+t)\,\xi\left(\frac{1}{1+t}\right) \qquad t \in [0,\pi] \tag{5.21}$$

then $\eta = (\eta(t))_{t\in[0\pi]}$ is also a continuous Gaussian process. Moreover, since

$$\mathbb{E}[\eta(t)] = 0, \quad \mathbb{E}[\eta(s)\eta(t)] = (1+t)(1+s)\mathbb{E}\left[\xi\left(\frac{1}{1+s}\right)\,\xi\left(\frac{1}{1+t}\right)\right] = \min(s,t), \tag{5.22}$$

η has the same finite dimensional distribution. Therefore, η is a Brownian motion.

5.2.1 Levy's construction

In this paragraph, we consider the construction of Brownian motion by Fourier series. This argument was done by N. Wiener and the following expression (5.23) is called **Wiener expansion**.

Similarly to (5.9), let ξ_n $(n = 0, 1, \cdots)$ be an independent random variables identically distributed as $\mathcal{N}(0, 1)$. We set

$$X_{n,t} := \frac{t}{\sqrt{\pi}}\,\xi_0 + \sqrt{\frac{2}{\pi}}\sum_{k=1}^{n}\frac{\sin(kt)}{k}\,\xi_k, \qquad Y_{n,t} := \sqrt{\frac{2}{\pi}}\sum_{k=n+1}^{\infty}\frac{\sin(kt)}{k}\,\xi_k \qquad t \in [0,\pi] \tag{5.23}$$

Then $X_t = X_{n,t} + Y_{n,t}$ $(t \in [0, \pi])$ is a Brownian motion. For fixed t,

$$\lim_{n\to\infty}\mathbb{E}[|X_{n,t} - X_t|^2] = 0 \tag{5.24}$$

Since $Y_{n,t}$ is also a continuous Gaussian process,

$$\begin{aligned}&\mathbb{E}[|Y_{n,t} - Y_{n,s}|^2] \le \mathbb{E}[|X_t - X_s|^2] = t - s,\\ &\mathbb{E}\big[|Y_{n,t} - Y_{n,s}|^4\big] = 3\Big(\mathbb{E}\big[|Y_{n,t} - Y_{n,s}|^2\big]\Big)^2 \le 3(t-s)^2 \qquad n = 1, 2, \cdots\end{aligned} \tag{5.25}$$

Therefore, since

$$M := \sup_n \mathbb{E}[Y_{n,0}^2] < \infty \tag{5.26}$$

and for positive numbers a, γ, δ,

$$\sup_n \mathbb{E}[|Y_{n,t} - Y_{n,s}|^\gamma] \le a\,|t-s|^{1+\delta} \tag{5.27}$$

hold, a subsequence of $\mathbf{Y}_n$ converges to 0. Thus $\mathbf{Y}_n$ converges to 0 in law. Therefore $\mathbf{Y}_n$ converges to 0 in probability, which implies that $\mathbf{X}_n$ converges to $\mathbf{X}$ in probability.

Moreover, since $\mathbf{X}_n$ is partial sum of independent $C([0,\pi];\mathbb{R})$-valued random variables, $\mathbf{X}_n$ converges to $\mathbf{X}$ almost surely. Hence, for almost every ω, $\mathbf{X}_{n,t} = \mathbf{X}_{n,t}(\omega)$ converges uniformly to $\mathbf{X}_t = \mathbf{X}_t(\omega)$ on $[0, \pi]$. In general, we can also prove almost sure convergence of

$$\sum_{k=0}^{n}\left(\int_0^t \phi_k(s)\,ds\right)\xi_k \tag{5.28}$$

for i.i.d. normal random variables ξ_k $(k = 0, 1, \cdots)$ and orthonormal basis on $L^2([0, T])$ ϕ_k $(k = 0, 1, \cdots)$ via Oottaviani's inequality. To be more precise, since for

$$\begin{aligned}&A_{n,k} := [(k-1)2^{-n}, k2^{-n}),\\ &\varphi_{n,k} := 2^{\frac{n-1}{2}}\mathbb{1}_{A_{n,k}} - 2^{\frac{n-1}{2}}\mathbb{1}_{A_{n,k+1}} \qquad k = 1, 3, \cdots, 2^n - 1,\ n = 1, 2, \cdots\end{aligned} \tag{5.29}$$

$\{1, \varphi_{n,k}; k = 1, 3, \cdots, 2^n - 1,\ n = 1, 2, \cdots\}$ is an complete orthonormal system on $L^2([0, 1])$,

$$X_t = t\xi_0 + \sum_{n=0}^{\infty}\sum_{k}\left(\int_0^t \varphi_{n,k}(s)\,ds\right)\xi_k \tag{5.30}$$

constructs a Brownian motion, which is called Levy's construction.

Question 5.6. Prove (5.24).

Question 5.7. Prove

$$\mathbb{E}\big[|Y_{n,t} - Y_{n,s}|^4\big] = 3\Big(\mathbb{E}\big[|Y_{n,t} - Y_{n,s}|^2\big]\Big)^2 \le 3(t-s)^2 \qquad n = 1, 2, \cdots \tag{5.31}$$

Hint: After choosing proper function f, use Parseval's identity

$$\frac{2}{\pi}\int_0^\pi |f(x)|^2 dx = \frac{a_0^2}{2} + \sum_{k=0}^{\infty} a_k^2 \tag{5.32}$$

for the Fourier cosine coefficients

$$a_k = \frac{2}{\pi}\int_0^\pi f(x)\,\cos kx\,dx. \tag{5.33}$$

5.3 Brownian motion

Through this section again we set $I = [0, \infty)$ or $[0, T]$ for some $T < \infty$.

Definition 5.8. Let $\mathbf{B} = (B_t)_{t\in I}$ be a one-dimensional stochastic process with $B_0 = 0$ satisfying that the increment $B_t - B_s$ $(s < t)$ is independent of $(B_\theta)_{\theta\le s}$ and its distribution is $N(0, t-s)$. Then the process is called **Brownian motion** or **Wiener process**.

Proposition 5.9. *If the stochastic process $\mathbf{X} = (X_t)_{t\in I}$ with $X_0 = 0$ has independent increments and its distribution is $N(0, t-s)$, then it is a Brownian motion, i.e. for $0 \le s_1 < t_1 \le s_2 < t_2 \le \cdots \le s_n < t_n$, $X_{t_k} - X_{s_k}$ $(k = 1, 2, \cdots, n)$ are independent random variables distributed by $N(0, t_k - s_k)$.*

Proof. First we remark that $\mathbf{X}$ is a Gaussian system. For any $0 = t_0 < t_1 < \cdots < t_n$, we set

$$\eta_k := X_{t_k} - X_{t_{k-1}} \qquad k = 1, 2, \cdots, n \tag{5.34}$$

which are independent random variable distributed by $N(0, t_k - t_{k-1})$. Since

$$X_{t_k} = \sum_{i=1}^{k} \eta_i, \tag{5.35}$$

$c_1 X_{t_1} + c_2 X_{t_2} + \cdots + c_n X_{t_n}$ is a linear combination of $\eta_1, \eta_2, \cdots, \eta_n$. Then $(X_{t_1}, X_{t_2}, \cdots, X_{t_n})$ is an n-dimensional Gaussian random variable, which means that $\mathbf{X}$ is also a Gaussian system. On the other hand, from the assumption of independent increments, for any $\theta \le s$,

$$\mathrm{Cov}(X_t - X_s, X_\theta) = \mathbb{E}[X_t - X_s]\,\mathbb{E}[X_\theta - X_0] = 0. \tag{5.36}$$

Therefore, $X_t - X_s$ is independent of $(B_\theta)_{\theta\le s}$, which implies Definition 5.8.

Assuming the condition in Definition 5.8, we compute the characteristic function of $\zeta_k := B_{t_k} - B_{s_k}$ $(k = 1, 2, \cdots, n)$. Since ζ_n is independent of $(B_s)_{s\le s_n}$ and its distribution is $N(0, t_n - s_n)$,

$$\begin{aligned}
&\mathbb{E}[\exp(ic_n\zeta_n + ic_{n-1}\zeta_{n-1} + \cdots + ic_1\zeta_1)] \\
&\quad = \mathbb{E}[\mathbb{E}[\exp(ic_n\zeta_n) \mid \sigma_{s_n}(\mathbf{B})]\exp(ic_{n-1}\zeta_{n-1} + \cdots + ic_1\zeta_1)] \\
&\quad = \exp\left(-\frac{c_n^2(t_n - s_n)}{2}\right)\mathbb{E}[\exp(ic_{n-1}\zeta_{n-1} + \cdots + ic_1\zeta_1)]
\end{aligned} \tag{5.37}$$

Repeatedly, we obtain

$$\mathbb{E}[\exp(ic_n\zeta_n + ic_{n-1}\zeta_{n-1} + \cdots + ic_1\zeta_1)] = \prod_{k=1}^{n} \exp\left(-\frac{c_k^2(t_k - s_k)}{2}\right) \tag{5.38}$$

Therefore, ζ_k $(k = 1, 2, \cdots, n)$ are independent random variables distributed by $N(0, t_k - s_k)$. ■

By Proposition 5.9, Brownian motion $\mathbf{B}$ is a Gaussian process. By the fourth moment property of Gaussian: $\mathbb{E}[|B_t - B_s|^4] = 3|t - s|^2$, which yields that Brownian motion is a continuous stochastic process (Check the definition).

Definition 5.10. Let $\mathbf{B}^{(1)}, \mathbf{B}^{(2)}, \cdots, \mathbf{B}^{(d)}$ be independent 1-dimensional Brownian motions.

(1). $\mathbf{B} = \{B_t = (B_t^{(1)}, B_t^{(2)}, \cdots, B_t^{(d)})\}_{t\in I}$ is called d**-dimensional Brownian motion**.

(2). for $x \in \mathbb{R}^d$, $\{B_t + x\}_{t\in I}$ is called d-dimensional Brownian motion from x.

(3). the distribution of d-dimensional Brown motion defined on $C(I; \mathbb{R}^d)$ is called d-dimensional **Wiener measure**. Especially in $d = 1$ the distribution is simply called **Wiener measure**.

The following is obvious:

Proposition 5.11. *Let $\{B_t\}_{t\in I}$ be a d-dimensional Brownian motion. Then for $s < t$, the increments $B_t - B_s$ are independent of $\{B_\theta\}_{\theta\le s}$ and its distribution is $N(0, (t-s)I_d)$, where I_d is a d-dimensional identity matrix.*

Example 5.12 (Linear combination of Brownian motions)**.** Let $\mathbf{B}^{(1)}, \mathbf{B}^{(2)}, \cdots, \mathbf{B}^{(d)}$ be independent 1-dimensional Brownian motions. We set for constants a_k $(k = 1, 2, \cdots, d)$,

$$X_t = \frac{1}{K}\sum_{k=1}^{d} a_k B_t^{(k)} \qquad K := \left(\sum_{k=1}^{d} a_k^2\right)^{\frac{1}{2}} > 0. \tag{5.39}$$

Then $\mathbf{X} = \{X_t\}_{t\in I}$ is a Gaussian process and satisfies

$$\mathbb{E}[X_t] = 0 \qquad \mathbb{E}[(X_t - X_s)^2] = t - s. \tag{5.40}$$

Moreover, for $\theta < s < t$,

$$\mathrm{Cov}(X_t - X_s, X_\theta) = \frac{1}{K^2}\sum_{i,j=1}^{d} a_i\, a_j\, \mathrm{Cov}(B_t^{(i)} - B_s^{(i)}, B_\theta^{(j)}) = 0, \tag{5.41}$$

which implies that $X_t - X_s$ is independent of $\{X_\theta\}_{\theta\le s}$. Therefore, $\mathbf{X}$ is also a 1-dimensional Brownian motion.

5.3.1 Donsker's invariance principle: functional CLT

In this section we consider an approximation of Brownian motion by random walk. The argument is known as a special case of functional central limit theorem (CLT). First we consider the pointwise convergence to Gaussian random variable, which is exactly the CLT.

Lemma 5.13 (The central limit theorem)**.** *Let ξ_j $(j = 1, 2, \cdots)$ be i.i.d. random variables satisfying*

$$\mathbb{E}[\xi_j] = 0, \qquad \mathbb{E}[\xi_j^2] = v \in (0, \infty), \qquad \mathbb{E}\big[|\xi_j|^3\big] < \infty. \tag{5.42}$$

We set

$$\eta_{n,t} := \frac{1}{\sqrt{nv}} \sum_{j=1}^{[nt]} \xi_j, \tag{5.43}$$

where $\eta_{n,t} = 0$ when $t \in [0, \frac{1}{n})$. Then for any $t \in I$, $\eta_{n,t}$ converges to a normal random variable with variance t in law.

Proof. We fix $t > 0$ and set $\eta_n = \eta_{n,t}$. The characteristic function of η_n is

$$\psi_n(z) := \mathbb{E}[\exp(iz\eta_n)] = \left(\varphi_\xi\left(\frac{z}{\sqrt{nv}}\right)\right)^{[nt]}, \tag{5.44}$$

where φ_ξ is the characteristic function of ξ. Since

$$\varphi_\xi\left(\frac{z}{\sqrt{nv}}\right) = 1 - \frac{z^2}{2n} + O(n^{-\frac{3}{2}}), \tag{5.45}$$

we have

$$\lim_{n\to\infty} \psi_n(z) = \lim_{n\to\infty} \left(1 - \frac{z^2}{2n} + O(n^{\frac{3}{2}})\right)^{[nt]} = \exp\left(\frac{z^2 t}{2}\right) \tag{5.46}$$

which is the characteristic function of $N(0, t)$. By Glivenko's theorem, the distribution of η_n converges weakly to $N(0, t)$. ■

The following proposition tells that the sequence of η converges to the points of Brownian motion.

Proposition 5.14. *For any d and $t_j \in I$ $(j = 1, 2, \cdots, d)$, as $n \to \infty$ $(\eta_{n,t_1}, \eta_{n,t_2}, \cdots, \eta_{n,t_d})$ converges to $(B_{t_1}, B_{t_2}, \cdots, B_{t_d})$ in law, where each entry B_{t_j} is a point of 1-dimensional Brownian motion.*

Proof. For $0 = t_0 < t_1 < t_2 < \cdots < t_d$, we set

$$Y_n^{(j)} := \eta_{n,t_j} - \eta_{n,t_{j-1}} \qquad (j = 1, 2, \cdots, d). \tag{5.47}$$

Then $Y_n^{(j)}$ are independent random variables. The characteristic function is computed as

$$\psi_n(z_1, z_2, \cdots, z_d) = \mathbb{E}\left[\exp\left(i \sum_{k=1}^{d} z_k\, Y_n^{(k)}\right)\right] = \prod_{k=1}^{d} \mathbb{E}\left[\exp\left(i z_k Y_n^{(k)}\right)\right] \tag{5.48}$$

By Lemma 5.13,

$$\mathbb{E}\left[\exp\left(i z_k Y_n^{(k)}\right)\right] \xrightarrow[n\to\infty]{} \exp\left(-\frac{1}{2} z_k^2 (t_k - t_{k-1})\right), \tag{5.49}$$

which implies that $(Y_n^{(1)}, Y_n^{(2)}, \cdots, Y_n^{(d)})$ converges to $(B_{t_1}, B_{t_2}, \cdots, B_{t_d})$ in law. ■

Remark 5.15. Since the path $\eta_{n,\cdot}$ is not continuous, the result in Proposition 5.14 never implies that $\eta_{n,\cdot}$ converges to Brownian motion in law. To guarantee that $\eta_{n,\cdot}$ becomes a continuous stochastic process, we modify Proposition 5.14 as follows.

Theorem 5.16 (Donsker's invariance principle)**.** *Let ξ_j $(j = 1, 2, \cdots)$ be i.i.d. random variables satisfying*

$$\mathbb{E}[\xi_j] = 0, \qquad \mathbb{E}[\xi_j^2] = v \in (0, \infty), \qquad \mathbb{E}\big[|\xi_j|^4\big] = cv^2. \tag{5.50}$$

We set

$$S_n := \sum_{j=1}^{n} \xi_j, \qquad S_0 = 0 \tag{5.51}$$

and define a stochastic process on $[0, 1]$ by

$$X_{n,t} = \frac{1}{\sqrt{nv}}\Big(S_{[nt]} + (nt - [nt])\xi_{[nt]+1}\Big). \tag{5.52}$$

Then $\mathbf{X}_n = (X_{n,t})_{t\in I}$ converges to Brownian motion in law.

Proof. We split the proof into two parts.

(I). We will prove

$$\mathbb{E}\Big[\big|X_{n,t} - X_{s,t}\big|^4\Big] \le 3c|t - s|^2 \tag{5.53}$$

Let $k := [nt]$ and $i := [ns]$. We set

$$I_1 := \sum_{j=i+2}^{k} \xi_j, \quad I_2 := (nt - k)\xi_{k+1}, \quad I_3 := (i + 1 - ns)\xi_{i+1} \tag{5.54}$$

Then we have

$$X_{n,t} - X_{n,s} = \begin{cases} \dfrac{1}{\sqrt{nv}}(I_1 + I_2 + I_3) & k \ge i + 1, \\ \dfrac{1}{\sqrt{nv}}\big(n(t - s)\xi_{k+1}\big) & k = i \end{cases}, \tag{5.55}$$

where $I_1 = 0$ when $k = i + 1$. We note here that I_1, I_2, I_3 are independent each other. Since $\mathbb{E}[I_j] = 0$,

$$\mathbb{E}\big[(I_1 + I_2 + I_3)^4\big] = \mathbb{E}\big[I_1^4\big] + \mathbb{E}\big[\big(I_2 + I_3\big)^4\big] + 6\mathbb{E}\big[I_1^2\big]\,\mathbb{E}\big[\big(I_2^2 + I_3^2\big)\big] \tag{5.56}$$

Since $nt - k$ and $i + 1 - ns$ are contained in $[0, 1]$ for $c \ge 1$,

$$\begin{aligned} &\mathbb{E}\big[I_1^4\big] = cv^2(k - i - 1) + 3v^2(k - i - 1)(k - i - 2) \le 3cv^2(k - i - 1)^2), \\ &\mathbb{E}\big[\big(I_2 + I_3\big)^4\big] \le 3v^2\big((nt - k) + (i + 1 - ns)\big)^2, \\ &\mathbb{E}\big[I_1^2\big]\,\mathbb{E}\big[\big(I_2^2 + I_3^2\big)\big] \le v^2(k - i - 1)\big((nt - k) + (i + 1 - ns)\big) \end{aligned} \tag{5.57}$$

and substituting them into (5.56), we obtain (5.53) for the case $k \ge i + 1$. For $k = i$, (5.53) is obvious.

(II). Let $0 \le s_1 < t_1 \le s_2 < t_2 \le \cdots \le s_d < t_d$. We set

$$Y_{n,t} := \frac{1}{\sqrt{nv}} S_{[nt]} \tag{5.58}$$

Then by Proposition 5.14, the distribution of $(Y_{n,t_1} - Y_{n,s_1}, Y_{n,t_2} - Y_{n,s_2}, \cdots, Y_{n,t_d} - Y_{n,s_d})$ converges to the distribution of $(B_{t_1} - B_{s_1}, B_{t_2} - B_{s_2}, \cdots, B_{t_d} - B_{s_d})$. On the other hand, the linear interpolate for each $X_{n,t} - X_{n,s} = \frac{1}{\sqrt{nv}}(I_2 + I_3)$ converges to 0 in L^2. Therefore, the distribution of $(X_{n,t_1} - X_{n,s_1}, X_{n,t_2} - X_{n,s_2}, \cdots, X_{n,t_d} - X_{n,s_d})$ converges to the distribution of $(B_{t_1} - B_{s_1}, B_{t_2} - B_{s_2}, \cdots, B_{t_d} - B_{s_d})$. By (5.53), we conclude that $\mathbf{X}_n$ converges to a Brownian motion in law. ■

Example 5.17 (Donsker)**.** Let X_{nt} be defined in (5.52). For bounded continuous function g defined on $\mathbb{R}^1$,

$$F(z) := g\Big(\sup_{0\le t\le 1} z(t)\Big) \tag{5.59}$$

is a bounded continuous function on $C([0,1])$. Then by Donsker's invariance principle,

$$\mathbb{E}\left[g\Big(\max_{0\le t\le 1} X_{n,t}\Big)\right] \xrightarrow[n\to\infty]{} \mathbb{E}\left[g\Big(\max_{0\le t\le 1} B_t\Big)\right] \tag{5.60}$$

and

$$\max_{0\le t\le 1} X_{n,t} = \frac{1}{\sqrt{nv}} \max_{0\le k\le n} S_k \tag{5.61}$$

Therefore, the maximum of random walk converges to the maximum of Brownian motion in law as $n\to\infty$. ■

The process (5.52) is an interpolate of $X_{n,\frac{k}{n}} = \frac{1}{\sqrt{nv}} S_k$ $(k = 0, 1, 2, \cdots, n)$. Thus it is not so difficult to imagine the path of the approximation $\mathbf{X}_n$ of Brownian motion is zigzag. The fluctuation of Brownian motion is predicted to be significantly large. The following proposition shows the estimate of **quadratic variations**.

Theorem 5.18 (Quadratic variation of Brownian motion)**.** *As $n\to\infty$,*

$$\sum_{2^n s\le k\le 2^n t} \left|B_{2^{-n}k} - B_{2^{-n}(k-1)}\right|^2 \to t - s \qquad a.s. \tag{5.62}$$

and

$$\mathbb{E}\left[\left|\sum_{2^n s\le k\le 2^n t} \left|B_{2^{-n}k} - B_{2^{-n}(k-1)}\right|^2 - (t-s)\right|^2\right] \to 0. \tag{5.63}$$

Proof. For simplicity, we take $s = 0$. Set

$$\xi_{n,k} := \big(B_{2^{-n}k} - B_{2^{-n}(k-1)}\big)^2 - 2^{-n}. \tag{5.64}$$

Then $\xi_{n,k}$ $(k = 1, 2, \cdots)$ are independent random variables with mean 0 and variance

$$\mathbb{E}\big[\xi_{n,k}^2\big] = \mathbb{E}\Big[\big(B_{2^{-n}k} - B_{2^{-n}(k-1)}\big)^4\Big] - 2^{-2n} = 2^{-2n+1} \tag{5.65}$$

Therefore, for

$$I_n := \sum_{k\ge 2^n t} \xi_{n,k} - (t - [2^n t]\, 2^{-n}) = \sum_{k\le 2^n t} \left|B_{2^{-n}k} - B_{2^{-n}(k-1)}\right|^2 - t, \tag{5.66}$$

we have the estimate

$$\mathbb{E}\big[\big|I_n\big|^2\big] = \sum_{k\le 2^n t} \mathbb{E}[\xi_{n,k}^2] + (t - [2^n t]\, 2^{-n})^2 \le 2^{-n}(2t+1), \tag{5.67}$$

which shows (5.62). From this estimate we have

$$\sum_{n=0}^{\infty} \mathbb{P}\big(I_n > 2^{-\frac{n}{4}}\big) \le \sum_{n=0}^{\infty} 2^{\frac{n}{2}}\, \mathbb{E}[I_n^2] < \infty \tag{5.68}$$

By the Borel–Cantelli lemma, we obtain (5.63). ■

Theorem 5.19 (Total variation of Brownian motion)**.** *For almost all path, the total variation on arbitrary interval is infinite.*

5.4 Compound Poisson process

Poisson process We consider Poisson process again. Let $S_0 = 0$ and $S_1 < S_2 < \cdots$ be the sequence of time when an event A occurs. Suppose that the increments $\eta_n = S_n - S_{n-1}$ $(n = 1, 2, \cdots)$ are independent and identically distributed by exponential distribution with parameter $\lambda > 0$. Then

$$S_n = \sum_{k=1}^{n} \eta_k \tag{5.69}$$

$\{S_n\}_{n=0,1,2,\cdots}$ is a random walk. Let N_t be the number of appearance of the event A until time t:

$$N_t = \min\{k \geq 0 \colon S_{k+1} > t\} = \sum_{n=1}^{\infty} \mathbb{1}_{[0,t]}(S_n) \tag{5.70}$$

Then the path $N_\cdot(\omega)$ is a right-continuous step function at $t = S_n(\omega)$ with jump width 1.

Definition 5.20 (Poisson process)**.** The stochastic process $\mathbf{N} = \{N_t\}_{t\geq 0}$ is called **Poisson process** with parameter $\lambda > 0$ if the following conditions are satisfied:

(1) $N_0 = 0$

(2) the process has independent increments and the distribution of $N_t - N_s$ is Poisson distribution with parameter $\lambda(t - s)$

(3) almost all paths are right-continuous step functions with jump width 1.

The construction above is from the view of stochastic models. In practice we generalize the Poisson processes below.

Compound Poisson process Let ξ_k $(k = 1, 2, \cdots)$ be square integrable i.i.d. random variables. We define a random walk $\mathbf{S}$ by

$$S_0 = 0, \qquad S_n = \sum_{k=1}^{n} \xi_k \quad (n = 1, 2, \cdots) \tag{5.71}$$

Let $\mathbf{N}$ be a Poisson process with parameter $\lambda > 0$ independent of $\mathbf{S}$. Then the stochastic process $\mathbf{X} = \{X_t\}_{t\geq 0}$ defined by

$$X_t = S_{N_t} = \sum_{k=1}^{N_t} \xi_k \tag{5.72}$$

is called **compound Poisson process**, where $X_t = 0$ when $N_t = 0$. Thus the time when the process $\mathbf{X}$ jumps denoted by θ is the same as that for $\mathbf{N}$ and the jump width is ξ_{N_θ}.

Proposition 5.21 (Property of compound Poisson process)**.** *Let $\mathbf{X} = \{X_t\}_{t\geq 0}$ be a compound Poisson process with parameter $\lambda > 0$, $\mathbb{E}[\xi_k] = a$ and $\mathbb{V}ar(\xi_k) = v < \infty$. Then*

- $\mathbb{E}[X_t] = a\lambda t$
- $\varphi(u, t) = \mathbb{E}[\exp(iuX_t)] = \exp(\lambda t\,(\varphi_\xi(u) - 1))$
- $\mathbf{X}$ *has independent increments and $X_t - X_s$ has the same distribution as X_{t-s}.*

Proof.

A Fourier Series Expansion

Taylor (Mclaurin) expansion gives an expression for a function in some class by a power series:

$$f(x) = \sum_{n=0}^{\infty} A_n\, x^n, \qquad A_n = \frac{f^{(n)}(0)}{n!}. \tag{A.73}$$

Fourier series expansion yields an expression for a **periodic** function by an infinite series of trigonometric functions (sin and cos):

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} \left(a_k\, \cos kx + b_k \sin kx\right), \tag{A.74}$$

where

$$a_0 = -\frac{1}{\pi} \int_{-\pi}^{\pi} f(t)\, dt, \quad a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t)\, \cos kt\, dt, \quad b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t)\, \sin kt\, dt. \tag{A.75}$$

For Taylor expansion, to show the expression of the coefficients A_n is not so difficult. However, it is not easy to show the expression of Fourier series expansion for the function f, i.e.,

$$S_n(x) := \frac{a_0}{2} + \sum_{k=1}^{n} \left(a_k\, \cos kx + b_k\, \sin kx\right). \tag{A.76}$$

The goal of this section is to prove the following theorem:

Theorem A.1 (Jordan-Lebesgue)**.** *Let f be a periodic function with 2π. If the function f is piecewise smooth, then*

$$\lim_{n\to\infty} S_n(x) = \frac{f(x+0) + f(x-0)}{2}. \tag{A.77}$$

Moreover, if the function f is piecewise continuous and smooth, then

$$\lim_{n\to\infty} S_n(x) = f(x). \tag{A.78}$$

We note here that the definitions of the terms "piecewise smooth" and "piecewise continuous" are the following.

Definition A.2 (Piecewise smooth)**.** The function f is said to be **piecewise smooth** on the interval $[a, b]$ when it satisfies the following conditions:

(1) except for finite points $x_1, x_2, \cdots, x_n \in [a, b]$, the function f is differentiable and the derivative f' is continuous

(2) at each $x_1, x_2, \cdots, x_n \in [a, b]$, there exist the left and right limits of f and f' and they are finite.

Definition A.3 (Piecewise continuous)**.** The function f is said to be **piecewise continuous** on the interval $[a, b]$ when it satisfies the following conditions:

(1) except for finite points $x_1, x_2, \cdots, x_n \in [a, b]$, the function f is continuous

(2) at each discontinuous points $x_1, x_2, \cdots, x_n \in [a, b]$, there exist the left and right limits of f and they are finite.

Example A.4. The ceiling function $f(x) = \lceil x \rceil$ is piecewise continuous.

Example A.5. For $n \in \mathbb{Z}$, the function

$$f(x) = \begin{cases} -1 & x \in \big((2n-1)\pi, 2n\pi\big] \\ 1 & x \in \big(2n\pi, (2n+1)\pi\big] \end{cases} \tag{A.79}$$

is piecewise smooth.

Example A.6. The periodic function $f(x) = \sqrt{1-x^2}$ $(-1 \le x \le 1)$ is piecewise continuous but not piecewise smooth. Since

$$f'(x) = -\frac{x}{\sqrt{1-x^2}}, \tag{A.80}$$

we have $f'(1+0) = -\infty$ and $f'(1-0) = -\infty$.

A.1 Dirichlet kernel

Before the proof of the Jordan-Lebesgue theorem, we will introduce Dirichlet kernel.

Definition A.7 (Dirichlet kernel)**.** For $n \in \mathbb{N}$ and $x \in \mathbb{R}$, Dirichlet kernel is defined by

$$D_n(x) = 1 + 2\sum_{k=1}^{n} \cos 2kx. \tag{A.81}$$

If we apply the Euler's formula, then Dirichlet kernel can be expressed as

$$D_n(x) = \sum_{k=-n}^{n} e^{-ikx}. \tag{A.82}$$

For the proof, see the following question.

Question A.8. Prove the equation (A.82).

Solution. We apply Euler's formula: $e^{ix} = \cos x + i \sin x$. Since

$$\cos kx = \frac{e^{i(kx)} + e^{-i(kx)}}{2}, \tag{A.83}$$

we have

$$\begin{aligned} D_n(x) &= 1 + \sum_{k=1}^{n} \left(e^{i(kx)} + e^{-i(kx)} \right) \\ &= 1 + \sum_{k=1}^{n} e^{i(kx)} + \sum_{k=1}^{n} e^{-i(kx)} \\ &= 1 + \sum_{k=1}^{n} e^{i(kx)} + \sum_{k=-n}^{-1} e^{i(kx)} = \sum_{k=-n}^{n} e^{-ikx}, \end{aligned} \tag{A.84}$$

which completes the proof. ■

Lemma A.9. *For $n \in \mathbb{N}$ and $x \in \mathbb{R} \setminus \{0\}$, Dirichlet kernel $D_n(x)$ can be expressed as*

$$D_n(x) = \frac{\sin\left(\frac{2n+1}{2}\right)x}{\sin\frac{x}{2}}. \tag{A.85}$$

We will skip the proof of this lemma (see the question below).

Question A.10. To show that

$$1 + 2\sum_{k=1}^{n} \cos 2kx = \frac{\sin(2n+1)x}{\sin x}. \tag{A.86}$$

Solution. For $n = 1$, by additive theorem for trigonometric functions,

$$\begin{aligned} \sin x\,(1 + 2\cos 2x) &= \sin x + 2\sin x \cos 2x \\ &= \sin x + \big(\sin 3x + \sin(-x)\big) = \sin x + \big(\sin 3x - \sin x\big) = \sin 3x \end{aligned} \tag{A.87}$$

For $n \in \mathbb{N}$, we assume that the statement is correct. Then for $n + 1$,

$$\begin{aligned} \sin x \left(1 + 2\sum_{k=1}^{n+1} \cos 2kx\right) &= \sin x \left(1 + 2\sum_{k=1}^{n} \cos 2kx + 2\cos 2(n+1)x\right) \\ &= \sin(2n+1)x + 2\cos 2(n+1)x \sin x \\ &= \sin(2n+1)x + \big(\sin(2n+3)x - \sin(2n+1)x\big) \\ &= \sin(2n+3)x. \end{aligned} \tag{A.88}$$

Therefore, by induction we complete the proof. ■

Question A.11. Show that

$$\frac{1}{\pi}\int_0^{\pi} \frac{\sin nx}{\tan\frac{x}{2}}\,dx = 1. \tag{A.89}$$

Solution. By Lemma A.9,

$$D_n(x) = \frac{\sin\left(\frac{2n+1}{2}\right)x}{\sin\frac{x}{2}} = \frac{\sin nx\cos\frac{x}{2} + \cos nx\sin\frac{x}{2}}{\sin\frac{x}{2}} = \cos nx + \frac{\sin nx}{\tan\frac{x}{2}}. \tag{A.90}$$

Therefore, since D_n is an even function with respect to the origin and $\displaystyle\int_{-\pi}^{\pi} D_n(x)\,dx = 2\pi$,

$$\int_0^{\pi} D_n(x)\,dx = \pi. \tag{A.91}$$

Therefore, since $\displaystyle\int_0^{\pi} \cos nx\,dx = 0$, we have $\displaystyle\frac{1}{\pi}\int_0^{\pi} \frac{\sin nx}{\tan\frac{x}{2}}\,dx = 1$. ■

A.2 Riemann-Lebesgue Theorem

Let f be a periodic function with 2π. As we see in the first part of this note, we define Fourier series expansion yields an expression for f by the following.

Definition A.12 (Fourier series expansion)**.**

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx)\,, \tag{A.92}$$

where

$$a_0 = -\frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt, \quad a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,\cos kt\,dt, \quad b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,\sin kt\,dt. \tag{A.93}$$

We also define the n-th partial sum of Fourier series by

$$S_n(x) = \frac{a_0}{2} + \sum_{k=1}^{n} (a_k \cos kx + b_k \sin kx)\,. \tag{A.94}$$

By the additive theorem of trigonometric functions, we have the following formulae: for different $m, n \in \mathbb{N}$,

$$\begin{aligned} &\int_0^{2\pi} \sin mx \cos nx\; dx = 0, \qquad \int_0^{2\pi} \sin mx \cos mx\; dx = 0, \\ &\int_0^{2\pi} \sin mx \sin nx\; dx = 0, \qquad \int_0^{2\pi} \cos mx \cos nx\; dx = 0. \end{aligned} \tag{A.95}$$

If the width of the range of integration is 2π, then the results above also holds. These formulae implies that the different trigonometric functions are orthogonal, which is the basic of the theory of the Fourier series expansion.

A.2.1 Bessel inequality and Parseval's identity

Let $(c_k)_{k\in\mathbb{N}_0}$ and $(d_k)_{k\in\mathbb{N}}$ be real-valued sequences. For $n \in \mathbb{N}$, $x \in \mathbb{R}$, we define

$$A_n(x) := \frac{c_0}{2} + \sum_{k=1}^{n} \big(c_k \cos kx + d_k \sin kx\big), \qquad \delta_n(x)^2 := \frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x) - A_n(x)|^2\; dx. \tag{A.96}$$

Then, we have the following theorem.

Proposition A.13. *$\delta_n(x)$ takes the minimum when*

$$c_0 = a_0, \quad c_k = a_k, \quad d_k = b_k \tag{A.97}$$

Proof. Let

$$B_n(x) = \sum_{k=1}^{n} \big(c_k \cos kx + d_k \sin kx\big). \tag{A.98}$$

Then,

$$\begin{aligned} \delta_n(x)^2 &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \Big|f(x) - \frac{c_0}{2} - B_n(x)\Big|^2\; dx \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \left(f(x)^2 + \frac{c_0^2}{4} + B_n(x)^2 - c_0\, f(x) + c_0\, B_n(x) - 2f(x)B_n(x)\right)\; dx \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)^2\,dx + \frac{c_0^2}{4} + \frac{1}{2\pi}\int_{-\pi}^{\pi} B_n(x)^2\,dx \\ &\qquad - \frac{c_0}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx + \frac{c_0}{2\pi}\int_{-\pi}^{\pi} B_n(x)\,dx - \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)B_n(x)\,dx \end{aligned} \tag{A.99}$$

Since

$$B_n(x)^2 = \sum_{k=1}^{n} \big(c_k \cos kx + d_k \sin kx\big)^2 + \sum_{k\neq j} \big(c_k \cos kx + d_k \sin kx\big)\big(c_j \cos jx + d_j \sin jx\big), \quad \text{(A.100)}$$

we have

$$\begin{aligned} \frac{1}{2\pi}\int_{-\pi}^{\pi} B_n(x)^2\,dx &= \frac{1}{2\pi}\sum_{k=1}^{n}\int_{-\pi}^{\pi}\big(c_k^2\cos^2 kx + d_k^2 \sin^2 kx\big)\,dx \\ &= \frac{1}{2\pi}\sum_{k=1}^{n}(c_k^2\pi + d_k^2\pi) \\ &= \frac{1}{2}\sum_{k=1}^{n}(c_k^2 + d_k^2). \end{aligned} \quad \text{(A.101)}$$

By periodicity of trigonometric functions,

$$\int_{-\pi}^{\pi} B_n(x)\,dx = \sum_{k=1}^{n}\left(c_k\int_{-\pi}^{\pi}\cos kx\,dx + d_k\int_{-\pi}^{\pi}\sin kx\,dx\right) = 0. \quad \text{(A.102)}$$

By the definition of a_k and b_k,

$$\begin{aligned} \int_{-\pi}^{\pi} f(x)B_n(x)\,dx &= \sum_{k=1}^{n}\left(c_k\int_{-\pi}^{\pi} f(x)\cos kx\,dx + d_k\int_{-\pi}^{\pi} f(x)\sin kx\,dx\right) \\ &= \pi\sum_{k=1}^{n}(a_k c_k + b_k d_k) \end{aligned} \quad \text{(A.103)}$$

Therefore, we obtain that

$$\begin{aligned} \delta_n(x)^2 &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)^2\,dx \\ &\qquad + \frac{1}{2}\left(\frac{c_0^2}{2} + \sum_{k=1}^{n}(c_k^2 + d_k^2) + c_0 a_0 - \sum_{k=1}^{n}(a_k c_k + b_k d_k)\right) \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)^2\,dx \\ &\qquad + \frac{1}{2}\left(\frac{(c_0 - a_0)^2}{2} + \sum_{k=1}^{n}\Big((c_k - a_k)^2 + (d_k - b_k)^2\Big)\right) - \frac{1}{2}\left(\frac{a_0^2}{2} + \sum_{k=1}^{n}(a_k^2 + b_k^2)\right) \end{aligned} \quad \text{(A.104)}$$

Therefore, $c_0 = a_0$, $c_k = a_k$ and $d_k = b_k$ yields the minimum of $\delta_n(x)$. ■

From the end part of proof of above theorem, since $\delta_n^2(x) \geq 0$, we have the inequality

$$\frac{a_0^2}{2} + \sum_{k=1}^{n}(a_k^2 + b_k^2) \leq \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)^2\,dx, \quad \text{(A.105)}$$

which is called **Bessel inequality**.

Corollary A.14 (Bessel inequality)**.** *Let f be a periodic function with 2π. Then, for any $n \in \mathbb{N}$,*

$$\frac{a_0^2}{2} + \sum_{k=1}^{n}(a_k^2 + b_k^2) \leq \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)^2\,dx, \quad \text{(A.106)}$$

If we have

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x) - S_n(x)|^2\,dx = 0, \quad \text{(A.107)}$$

then, for any $n \in \mathbb{N}$, the equality holds in Bessel inequality. Therefore, we obtain

$$\frac{a_0^2}{2} + \sum_{k=1}^{\infty}(a_k^2 + b_k^2) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)^2\,dx, \tag{A.108}$$

which is called **Parseval's identity**.

Remark A.15. The condition

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x) - S_n(x)|^2\,dx = 0 \tag{A.109}$$

implies that for any $\varepsilon > 0$ $\sup_{x\in\mathbb{R}} |f(x) - S_n(x)| < \varepsilon$, i.e., f is continuous, integrable and S_n converges uniformly to f. In the case that f is piecewise continuous, we define f at discontinuous point x_0 by

$$\frac{f(x_0+0) + f(x_0-0)}{2}. \tag{A.110}$$

Then, $S_n(x)$ converges uniformly to f and Parseval's identity holds.

From Bessel inequality, we conclude the following theorem.

Theorem A.16 (Riemann-Lebesgue)**.** *Let f be a periodic function with 2π. If the function f is piecewise continuous, then*

$$\begin{aligned} a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \xrightarrow[n\to\infty]{} 0 \\ b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \xrightarrow[n\to\infty]{} 0 \end{aligned} \tag{A.111}$$

Proof. We give another proof here. Since the function $f(x)\sin nx$ is periodic with respect to 2π, we consider

$$\int_0^{2\pi} f(x)\sin nx\,dx = \sum_{k=0}^{n-1}\left(\int_{\frac{2k}{n}\pi}^{\frac{2k+1}{n}\pi} + \int_{\frac{2k+1}{n}\pi}^{\frac{2(k+1)}{n}\pi}\right) f(x)\sin nx\,dx. \tag{A.112}$$

We take $x = y + \frac{\pi}{n}$. Then, the second integral becomes

$$\int_{\frac{2k+1}{n}\pi}^{\frac{2(k+1)}{n}\pi} f(x)\sin nx\,dx = \int_{\frac{2k}{n}\pi}^{\frac{2k+1}{n}\pi} f(y+\tfrac{\pi}{n})\sin(ny+\pi)\,dy = -\int_{\frac{2k}{n}\pi}^{\frac{2k+1}{n}\pi} f(y+\tfrac{\pi}{n})\sin ny\,dy. \tag{A.113}$$

Therefore, we obtain that

$$\int_0^{2\pi} f(x)\sin nx\,dx = \sum_{k=0}^{n-1}\int_{\frac{2k}{n}\pi}^{\frac{2k+1}{n}\pi} \big(f(x) - f(x+\tfrac{\pi}{n})\big)\sin nx\,dx. \tag{A.114}$$

Now we can estimate

$$\left|\int_0^{2\pi} f(x)\sin nx\,dx\right| \le \sum_{k=0}^{n-1}\int_{\frac{2k}{n}\pi}^{\frac{2k+1}{n}\pi} \big|f(x) - f(x+\tfrac{\pi}{n})\big|\,dx = \int_0^{2\pi} \big|f(x) - f(x+\tfrac{\pi}{n})\big|\,dx. \tag{A.115}$$

Since $f(y+\frac{\pi}{n})$ converges uniformly to $f(x)$ on $[0, 2\pi]$, the rightmost side implies that

$$\lim_{n\to\infty}\int_0^{2\pi} \big|f(x) - f(x+\tfrac{\pi}{n})\big|\,dx = \int_0^{2\pi} \lim_{n\to\infty}\big|f(x) - f(x+\tfrac{\pi}{n})\big|\,dx = 0, \tag{A.116}$$

which completes the proof. ∎

A.3 Proof of Jordan-Lebesgue theorem

We will change the expression of n-th partial sum of Fourier series expansion. By the definition of Dirichlet kernel and its properties, we have

$$\begin{aligned} S_n(x) &= \frac{a_0}{2} + \sum_{k=1}^{n} (a_k \cos kx + b_k \sin kx) \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\,dt + \frac{1}{\pi}\sum_{k=1}^{n}\int_{-\pi}^{\pi} f(x)\,(\cos kt\cos kx + \sin kt \sin kx)\,dt \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\left(1 + 2\sum_{k=1}^{n}\cos k(x-t)\right)dt \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\frac{\sin\frac{(2n+1)(x-t)}{2}}{\sin\frac{x-t}{2}}\,dt \end{aligned} \tag{A.117}$$

We take $t = x + 2\xi$. Then,

$$S_n(x) = \frac{1}{2\pi}\int_{-\frac{x+\pi}{2}}^{-\frac{x-\pi}{2}} f(x+2\xi)\,\frac{\sin(2n+1)\xi}{\sin\xi}\,d\xi. \tag{A.118}$$

By periodicity of f, $f(x+2\xi)$ is also periodic with π (variable is ξ). Since

$$\frac{\sin(2n+1)\xi}{\sin\xi} = \frac{\sin(n\xi\cos\xi}{\sin\xi} + \frac{\cos 2n\xi\sin\xi}{\sin\xi} = \sin 2n\xi\cot\xi + \cos 2n\xi, \tag{A.119}$$

the integrand is periodic with π. Therefore, since the integral of the periodic function never change its value and

$$\begin{aligned} \int_{-\frac{\pi}{2}}^{0} f(x+2\xi)\,\frac{\sin(2n+1)\xi}{\sin\xi}\,d\xi &= \int_{\frac{\pi}{2}}^{0} f(x+2(-\eta))\frac{\sin(2n+1)(-\eta)}{\sin(-\eta)}\,(-d\eta) \\ &= \int_{0}^{\frac{\pi}{2}} f(x-2\eta)\frac{\sin(2n+1)\eta}{\sin\eta}\,d\eta, \end{aligned} \tag{A.120}$$

we have

$$S_n(x) = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(x+2\xi)\,\frac{\sin(2n+1)\xi}{\sin\xi}\,d\xi = \frac{1}{\pi}\int_{0}^{\frac{\pi}{2}} (f(x+2\xi)+f(x-2\xi))\,\frac{\sin(2n+1)\xi}{\sin\xi}\,d\xi. \tag{A.121}$$

Since

$$\int_{0}^{\frac{\pi}{2}} \frac{\sin(2n+1)\xi}{\sin\xi}\,d\xi = \int_{0}^{\frac{\pi}{2}}\left(1 + 2\sum_{k=1}^{n}\cos 2k\xi\right)d\xi = \frac{\pi}{2}, \tag{A.122}$$

we can express $f(x)$ by

$$f(x) = \frac{1}{\pi}\,\frac{\pi}{2}\,(2f(x)) = \frac{1}{\pi}\int_{0}^{\frac{\pi}{2}} \frac{\sin(2n+1)\xi}{\sin\xi}\,(2f(x))\,d\xi \tag{A.123}$$

Note here that $\cos 2k\xi$ is symmetric with respect to $\xi = \frac{\pi}{4}$. Therefore, we obtain that

$$\pi\Big(S_n(x) - f(x)\Big) = \int_{0}^{\frac{\pi}{2}} \Big(f(x+2\xi) + f(x-2\xi) - 2f(x)\Big)\,\frac{\sin(2n+1)\xi}{\sin\xi}\,d\xi. \tag{A.124}$$

Let

$$g(\xi) := \frac{f(x+2\xi) + f(x-2\xi) - 2f(x)}{\sin\xi}. \tag{A.125}$$

Then, the equation (A.124) can be written as

$$\pi\Big(S_n(x) - f(x)\Big) = \int_0^{\frac{\pi}{2}} g(\xi)\, \sin(2n+1)\xi \, d\xi. \tag{A.126}$$

If the function g is piecewise continuous, then by Riemann-Lebesgue theorem,

$$\lim_{n\to\infty} \Big(S_n(x) - f(x)\Big) = \lim_{n\to\infty} \frac{1}{\pi} \int_0^{\frac{\pi}{2}} g(\xi)\, \sin(2n+1)\xi \, d\xi = 0. \tag{A.127}$$

So we need to prove the (piecewise) continuity of g.

Since $\sin\xi = 0$ when $\xi = 0$, we consider continuity as $\xi \to 0$. We take $t = 2\xi$. Then,

$$\begin{aligned} \lim_{\xi\to 0} g(\xi) &= \lim_{t\to 0} \left(\frac{f(x+t) - f(x)}{\sin\frac{t}{2}} + \frac{f(x-t) - f(x)}{\sin\frac{t}{2}} \right) \\ &= \lim_{t\to 0} \left(\frac{f(x+t) - f(x)}{t} + \frac{f(x-t) - f(x)}{t} \right) \cdot \frac{\frac{t}{2}}{\sin\frac{t}{2}} \cdot 2 \\ &= \Big(f'(x) + (-f'(x))\Big) \cdot 1 \cdot 2 = 0 \end{aligned} \tag{A.128}$$

Therefore, since f is differentiable, by Riemann-Lebesgue theorem,

$$\lim_{n\to\infty} \Big(S_n(x) - f(x)\Big) = 0, \tag{A.129}$$

that is, $S_n \to f(x)$ as $n \to \infty$, which completes the proof of Jordan-Lebesgue theorem.

Example A.17. The periodic function with respect to 2π satisfying

$$f(x) = \begin{cases} 0 & -\pi \le x < 0 \\ x & 0 \le x < \pi \end{cases} \tag{A.130}$$

We have discontinuity at $x = 0$ and $x = \pi$. Since at $x = 0, \pi$,

$$f(x+0) \neq \infty, \quad f(x-0) \neq \infty, \quad f'(x+0) \neq \infty, \quad f'(x-0) \neq \infty, \tag{A.131}$$

the function f is piecewise continuous and smooth. By the Jorda-Lebesgue theorem, the Fourier series expansion of f converges to

$$\frac{f(0+0) + f(0-0)}{2} = 0, \quad \frac{f(\pi+0) + f(\pi-0)}{2} = \frac{\pi}{2} \tag{A.132}$$

at $x = 0, \pi$, respectively. Indeed, we can check by simulation that

$$S_n(x) = \frac{\pi}{4} + \sum_{k=1}^{n} \left(-\frac{1 + (-1)^{k-1}}{k^2\pi} \cos kx + \frac{(-1)^{k-1}}{k} \sin kx \right) \tag{A.133}$$

converges to $f(x)$. ■

Example A.18. The periodic function with respect to 2 satisfying for $-1 < x \le 1$,

$$f(x) = \sqrt{1 - x^2} \tag{A.134}$$

We have discontinuity at $x = 1$. Since at $x = 1$,

$$f(1+0) = 0, \quad f(1-0) = 0, \tag{A.135}$$

the function f is piecewise continuous. However, since $f'(x) = -\frac{x}{\sqrt{1-x^2}}$, at $x = 1$,

$$f'(1+0) = -\infty, \quad f'(1-0) = -\infty, \tag{A.136}$$

which implies that the function f is not piecewise smooth. Therefore, we cannot apply the Jordan-Lebesgue theorem, i.e., we cannot say that the Fourier series expansion of f converges to $\frac{f(x+0)+f(x-0)}{2}$. The Fourier series expansion of f can be formally expressed by

$$S_n(x) = \frac{\pi}{4} + \sum_{k=1}^{\infty} \frac{J_1(n\pi)}{k} \cos k\pi x \tag{A.137}$$

where $J_1(x)$ is called the 1st-degree **Bessel function**. The nth partial sum of Fourier series actually satisfies that $S_n(1) \to 0$ as $n \to \infty$. This is a special case. ■